



1 Boundary Conditions 1

Which component of the electric field intensity is always continuous at the boundary? Choose between the *tangential*, *normal*, *horizontal* or *vertical* component and explain why.

2 Boundary Conditions 2

The normal component of which quantity is discontinuous at the boundary when the free charge density $\sigma_f > 0$? Choose between $\mathbf{E}$, $\mathbf{D}$, $\mathbf{H}$ or $\mathbf{B}$ and explain why.

3 Radiated power through a boundary s

Consider the power radiated through a boundary s as expressed below and presented in the lectures. Explain the integration terms. How much of the energy is spent on heating due to the Ohmic loss?

$$\oint_S [\mathbf{E} \times \mathbf{H}] \, ds = - \int_V \left(\mathbf{H} \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \frac{\partial \mathbf{D}}{\partial t} \right) dv - \int_V \sigma \mathbf{E}^2 dv - \int_V \mathbf{J}^{(\text{source})} \mathbf{E} dv \quad (1)$$

4 Power radiated from a spherical surface

Problem 9.35 (*Introduction to Electromagnetism by Griffiths*, 4th Edition):

Suppose

$$\mathbf{E}(r, \theta, \phi, t) = A \frac{\sin \theta}{r} \left[\cos(kr - \omega t) - \frac{1}{kr} \sin(kr - \omega t) \right] \hat{\phi} \quad \text{with } \frac{\omega}{k} = c \quad (2)$$

(This is, incidentally, the simplest possible *spherical wave*. For notational convenience, let $(kr - \omega t) \equiv u$ in your calculations.)

- a) Show that $\mathbf{E}$ obeys all four of Maxwell's equations, in vacuum, and find the associated magnetic field.
- b) Calculate the Poynting vector. Average $\mathbf{S}^1$ over a full cycle to get the intensity vector $\mathbf{I}$. (Does it point in the expected direction? Does it fall off like r^{-2} , as it should?)

¹The variable $\mathbf{P}$ was used in the lecture for the Poynting vector, but different literature is using $\mathbf{S}$ equivalently.

- c) Integrate $\mathbf{I} d\mathbf{a}$ over a spherical surface to determine the total power radiated.
[Answer: $\frac{4\pi A^2}{3\mu_0 c}$]

5 Polarization

A plane wave propagating in the z -direction is given by $\mathbf{E}(z) = \mathbf{E}_0 \exp(-jkz)$ (in the frequency domain). In this problem, note that also straight lines and circles are special cases of an ellipse.

- a) Let $\mathbf{E}_0 = E_0 \hat{\mathbf{y}}$. Draw the polarization ellipse of the electric field. Also draw the polarization ellipse of the magnetic field.
- b) Let $\mathbf{E}_0 = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$. Draw the polarization ellipse of the electric field.
- c) Let $\mathbf{E}_0 = E_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})$. Draw the polarization ellipse of the electric field.
- d) What is birefringence?